

SQL Constraints and Key Selection

Authored by: Arpit Raj, LNMIIT Jaipur

Primary Key Selection Rules

Rule 1 — Must be Unique: Every row should have a different value.

Rule 2 — Must Never Be NULL: Primary Keys cannot contain NULL values.

Rule 3 — Should Be Stable: Changing a Primary Key affects all related Foreign Keys. Stable keys reduce maintenance.

Rule 4 — Keep It Small: Smaller keys mean Smaller indexes, Faster joins, Lower storage requirements.

Rule 5 — Avoid Business Information: Business rules change. Keys should not. (Example: Avoid EmployeeEmail, Prefer EmployeeID).

Rule 6 — Prefer Single Column: Single-column Primary Keys are easier to reference and index. Composite Primary Keys should be used only when the natural identity truly requires multiple columns.

Key Constraints in SQL

PRIMARY KEY Constraint

A Primary Key constraint ensures: uniqueness, NOT NULL, row identification.

Column-Level Syntax

```
CREATE TABLE Student
(
    StudentID INT PRIMARY KEY,
    Name VARCHAR(50)
);
```

Table-Level Syntax

```
CREATE TABLE Student
(
    StudentID INT,
    Name VARCHAR(50),
    PRIMARY KEY(StudentID)
);
```

Composite Primary Key

```
CREATE TABLE Enrollment
(
    StudentID INT,
    CourseID INT,
    PRIMARY KEY(StudentID, CourseID)
);
```

FOREIGN KEY Constraint

A Foreign Key creates a relationship between two tables.

```
CREATE TABLE Department
(
    DepartmentID INT PRIMARY KEY,
    DepartmentName VARCHAR(50)
);

CREATE TABLE Employee
(
    EmployeeID INT PRIMARY KEY,
    DepartmentID INT,
    FOREIGN KEY (DepartmentID)
    REFERENCES Department(DepartmentID)
);
```

UNIQUE Constraint

Unique constraint prevents duplicate values.

```
CREATE TABLE Employee
(
    EmployeeID INT PRIMARY KEY,
    Email VARCHAR(100) UNIQUE
);

/* Composite UNIQUE */
CREATE TABLE Student
(
    RollNo INT,
    Semester INT,
    UNIQUE(RollNo, Semester)
);
```

Managing Constraints (ALTER TABLE)

Sometimes, tables already exist. You need to add constraints later.

Add Primary Key

```
ALTER TABLE Student
ADD PRIMARY KEY(StudentID);
```

Can a PRIMARY KEY be added later? Yes. Use ALTER TABLE. However, existing data must satisfy the rules (no duplicate values, no NULL values).

```
ALTER TABLE Student
ADD CONSTRAINT PK_Student
PRIMARY KEY(StudentID);
```

REFERENCES Clause

The REFERENCES keyword tells SQL which table and which column is being referenced.

```
FOREIGN KEY(CustomerID)
REFERENCES Customer(CustomerID)
```

Meaning: CustomerID in this table must already exist in Customer.

Removing a Foreign Key

A Foreign Key constraint can be removed using ALTER TABLE with the appropriate DROP CONSTRAINT.

```
ALTER TABLE Employee
DROP CONSTRAINT FK_Department;
```